frontiers in Physics

ORIGINAL RESEARCH
published: 12 July 2019
doi: 10.3389/fphy.2019.00097

Effect of Grain Boundary on Diffusion of P in Alpha-Fe: A Molecular Dynamics Study

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In this study, we have investigated the effect of the grain boundary (GB) on the diffusion of a Phosphorus (P) atom in alpha-Fe using molecular dynamics simulations. A Fe-P mixed <110> dumbbell is created in the six symmetric tilt grain boundary (STGB) models. The dumbbells are allowed to migrate at different temperatures from 400 to 1,000 K, with starting positions between 5 to 10Å away from the GB core. The trajectories and mean square displacements (MSD) have been recorded to analyze the diffusion details. The Nudged Elastic Band (NEB) method has been used to study the energy barrier at different positions around the GBs. Our simulation results demonstrate that the GB structure affects the diffusion mechanisms of Fe-P dumbbell. The two low Σ favored GBs display significantly weak trapping effect, which is consistent with the formation energy distribution. The reduction in the migration barrier has been observed due to the decrease of distance from the GB center. Furthermore, the barriers of migration toward the GB are lower than the barriers of migration away from the GB. As evident by NEB calculation, absorption sink effect of GB has been observed. This effect saturates as the distance reaches 8Å or more. Our simulation results provide an insight into the GB trapping effect in alpha-Fe.

Keywords: grain boundary, phosphorus diffusion, alpha-iron, NEB, trapping effect

OPEN ACCESS

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Specialty section:
This article was submitted to Computational Physics, a section of the journal Frontiers in Physics

Received: *05 March 2019*
Accepted: *21 June 2019*
Published: *12 July 2019*

Citation:
Azeem MM, Wang Q, Zhang Y, Liu S and Zubair M (2019) Effect of Grain Boundary on Diffusion of P in Alpha-Fe: A Molecular Dynamics Study. Front. Phys. 7:97. doi: 10.3389/fphy.2019.00097

INTRODUCTION

Typically, structural materials in the nuclear industry are polycrystalline materials having pre-existent structural impurities. The grain boundaries (GBs) substantially influence mechanical properties of polycrystalline materials [1]. A small concentration of any impurity can modify the GBs diffusion rate during crystal growth [2, 3]. Hence, understanding the behavior of impurities and their interaction with the lattice defects and the grain boundaries has become crucial for designing a better engineering material [4–8].

The diffusion phenomenon of the defects and solute elements around the GBs has long been an area of intrigue for the researchers due to its impact on various mechanical properties, such as; corrosion resistance, and embrittlement [9]. The tempered embrittlement and radiation embrittlement has been observed in steel during rapid cooling from higher temperature and irradiation [10]. Phosphorus is one of the major impurity amongst the embrittlement species in steel [10–12]. The Phosphorus grain boundary segregation plays an important role in the intergranular embrittlement of the reactor pressure vessel (RPV), especially under irradiation